

1 Stat Theorems

$$E\{E[Y|X]\} = E[Y]$$

$$E\{var[Y|X]\} + Var(E[Y|X]) = Var(Y).$$

For a non-negative RV X and $a > 0$, by **Markov's Inequality**,

$$P(X \geq a) \leq \frac{E(X)}{a}.$$

For RV X with finite $\sigma^2 \neq 0$ (hence finite μ) and $k > 0$, by **Chebyshev's Inequality**,

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

For RV X and any convex function f ($f''(x) \geq 0$), by **Jensen's Inequality**,

$$E(f(X)) \geq f(E(X)).$$

with equality only with g is a constant function or affine, or almost surely constant X .

2 Probability

2.1 Random Variables

A **random variable** maps the sample space to a measurable space, typically a subset of the reals:

$$X: \Omega \rightarrow \mathbb{R}$$

Interpreting RVs arisen from **random experiments**, their outcomes or data are obtained as **realisations**. In general, we denote RVs with capital letters, and its realisations small letters.

Distribution of the RV

Obtaining r realisations and creating a histogram, as $r \rightarrow \infty$, the limiting histogram converges to the **distribution** of the RV.

2.2 Summary Statistics

Summary of Sample

Given $x_1, \dots, x_n$ realisations of a real-valued RV, we have the **mean**, **mean-square**, **variance** and **standard deviation**:

$$\bar{x} := \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{x}^2 := \frac{1}{n} \sum_{i=1}^n x_i^2$$

$$v := \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad SD := \sqrt{v}$$

We say $x_1, \dots, x_n$ are **centred at $\bar{x}$** and the **spread** is SD . We have, algebraically, the result $v = \bar{x}^2 - \bar{x}^2$.

Summary of Random Variable

Given RV X , we have the **expectation** and **variance**:

$$E(X) := \begin{cases} \sum_{i=1}^k x_i P(X = x_i) & \text{discrete} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{continuous} \end{cases}$$

$$Var(X) := E\{(X - \mu)^2\} = E(X^2) - E(X)^2.$$

The **standard deviation** is $SD(X) := \sqrt{Var(X)}$. For any $h: \mathbb{R} \rightarrow \mathbb{R}$, we have:

$$E\{h(X)\} := \begin{cases} \sum_{i=1}^k h(x_i) P(X = x_i) & \text{discrete} \\ \int_{-\infty}^{\infty} h(x) f(x) dx & \text{continuous} \end{cases}$$

With the **linearity of expectation**, given RVs X, Y and constants a, b, c :

$$E(aX + bY + c) = aE(X) + bE(Y) + c.$$

Law of Large Numbers

For $x_1, \dots, x_r$ realisations from X ,

$$\frac{1}{r} \sum_{i=1}^r h(x_i) = E\{h(X)\}$$

In particular, $\bar{x} \rightarrow E(X)$ and $SD_{\bar{x}} \rightarrow SD(X)$. The law of large numbers motivate **Monte Carlo approximations**, which approximates $E\{h(X)\}$ with large number of realisations.

2.3 Distributions

Bernoulli

Given $X \sim \text{Bernoulli}(p)$ for some $p \in [0, 1]$,

$$p_X(x) = \begin{cases} p & x = 1 \\ 1 - p & x = 0 \end{cases}$$

$$E(X) = p, \quad Var(X) = p(1 - p).$$

X has probability of success (of being 1) p .

Binomial

Given $X \sim \text{Bin}(n, p)$ for some $n \in \mathbb{N}, p \in [0, 1]$,

$$p_X(x) = \binom{n}{x} p^x (1 - p)^{n-x}, \quad k \in \mathbb{R}^+.$$

$$E(X) = np, \quad Var(X) = np(1 - p).$$

X is the sum of n IID RVs $X_1, \dots, X_n \sim \text{Bernoulli}(p)$.

$$Y \sim (m, p) \implies X + Y \sim \text{Bin}(n + m, p).$$

$$X \sim N(np, np(1 - p)) \text{ as } n \rightarrow \infty.$$

Multinomial

Let $\Omega = \{\|x\|_1 = n : x \in \mathbb{N}^k\}$. Then given $\mathbf{X} \sim \text{Multinomial}(n, \mathbf{p})$ for some $n \in \mathbb{N}, \mathbf{p} \in [0, 1]^k$, such that $\|\mathbf{p}\|_1 = 1$,

$$p_{\mathbf{X}}(\mathbf{x}) = \binom{n}{x_1, x_2, \dots, x_k} \prod_{i=1}^k p_i^{x_i}.$$

$$E(X_i) = np_i, \quad Var(X_i) = np_i(1 - p_i).$$

$$Cov(X_i, X_j) = -np_i p_j, \quad 1 \leq i, j \leq k, i \neq j.$$

Normal

Given $X \sim N(\mu, \sigma^2)$ for some $\mu \in \mathbb{R}, \sigma > 0$,

$$f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left\{-\frac{(x - \mu)^2}{2\sigma^2}\right\}, \quad x \in \mathbb{R}$$

In particular, the **standard normal** RV $Z \sim N(0, 1)$ has PDF

$$\phi(x) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{x^2}{2}\right\}, \quad x \in \mathbb{R}$$

$$X \sim N(\mu, \sigma^2) \implies \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Gamma

Given $X \sim \text{Gamma}(\alpha, \lambda)$ for some $\alpha > 0, \lambda > 0$,

$$f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}, \quad x \in \mathbb{R}^+$$

where $\Gamma(x) = \int_0^\infty u^{x-1} e^{-u} du$.

$$E(X) = \frac{\alpha}{\lambda}, \quad Var(X) = \frac{\alpha}{\lambda^2}.$$

$\Gamma(x) = (x - 1)\Gamma(x - 1)$ is the number such that the density integrates to 1.

$X_i \sim G(\alpha_i, \lambda) \implies \sum_{i=1}^n X_i \sim \text{Gamma}\left(\sum_{i=1}^n \alpha_i, \lambda\right)$.

$$\text{Exponential}(\lambda) = \text{Gamma}(1, \lambda).$$

Chi-Square

Given $X \sim \chi_k^2$ for some $k \in \mathbb{N}^*$,

$$f_X(x) = \frac{1}{2^{k/2} \Gamma(k/2)} x^{(k/2)-1} e^{-x/2}, \quad x > 0.$$

$$E(X) = k, \quad Var(X) = 2k.$$

$$Z \sim N(0, 1) \implies Z^2 \sim \chi_1^2 \text{ and } X \sim \text{Gamma}\left(\frac{k}{2}, \frac{1}{2}\right)$$

$$X = \sum_{i=1}^k X_i, \text{ for } X_1, X_2, \dots, X_k \text{ IID } \chi_1^2 \text{ RVs.}$$

t Distribution

Given $X \sim t_n$ for some $n \in \mathbb{R}$ (almost always $n \in \mathbb{Z}$),

$$f_X(x) = \frac{\Gamma\left(\frac{n+1}{2}\right)}{\sqrt{\pi n} \Gamma\left(\frac{n}{2}\right)} \left(1 + \frac{x^2}{n}\right)^{-\frac{n+1}{2}}.$$

$$E(X) = 0, \quad Var(X) = \begin{cases} \frac{n}{n-2} & n > 2 \\ \infty & 1 < n \leq 2 \end{cases}.$$

For independent $Z \sim N(0, 1)$ and $V \sim \chi_n^2$,

$$t_n = \frac{Z}{\sqrt{V/n}}.$$

As $n \rightarrow \infty$, the $t_n \rightarrow N(0, 1)$

2.4 Multi-RVs

Given two RVs X, Y , we have the **joint PDF** $f: \mathbb{R}^2 \rightarrow [0, \infty)$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1,$$

and the **marginal density** of X (similarly for $f_Y(y)$)

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy, \quad x \in \mathbb{R}.$$

For $h: \mathbb{R}^2 \rightarrow \mathbb{R}$,

$$E\{h(X, Y)\} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f(x, y) dx dy$$

Covariance

We define the covariance between X, Y

$$Cov(X, Y) := E\{(X - \mu_X)(Y - \mu_Y)\} = E(XY) - \mu_X \mu_Y$$

We have the results:

$$Var(X) = Cov(X, X).$$

$$Cov\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j Cov(X_i, Y_j)$$

Independence $\iff \forall x, y \quad f_{X|Y}(x|y) = f_X(x)$

$$\iff \forall A_1, \dots, A_n \subseteq \Omega$$

$$P(X_1 \in A_1, \dots, X_n \in A_n) = \prod_{i=1}^n P(X_i \in A_i)$$

$$\iff \forall s, t \quad M_{XY}(s, t) = M_X(s) M_Y(t)$$

$$\implies E[g(X)h(Y)] = E[g(X)]E[h(Y)]$$

$$\implies Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i)$$

$$\implies Cov(X_i, Y) = 0 \quad \text{Uncorrelated RVs.}$$

Normal $X, Y \wedge Cov(X, Y) = 0 \implies$ Independence

Independent and Identically Distributed RVs

Given IID RVs $X_1, \dots, X_n$ of expectation μ and variance σ^2 , we have the **expectation and variance** of the sum $S_n = \sum_{i=1}^n X_i$:

$$E(S_n) = n\mu \quad Var(S_n) = n\sigma^2.$$

In addition, we define the **sample variance**

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

$$E(S^2) = \sigma^2, \quad Var(S^2) = \frac{2\sigma^4}{n-1}.$$

By the **Central Limit Theorem**, as $n \rightarrow \infty$,

$$S_n \sim N(n\mu, n\sigma^2).$$

IID Normal Rvs

The **expectation and variance** of the sample mean $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$:

$$E(\bar{X}) = \mu, \quad Var(\bar{X}) = \frac{\sigma^2}{n}.$$

With the following results:

- $\frac{(n-1)S^2}{\sigma^2} = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} \sim \chi_{n-1}^2$
- $\bar{X}$ and S^2 are independent.

We have

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}.$$

Random Vector

For $\mathbf{X} = (X_1, \dots, X_k)$ a random vector of k RVs, we define

$$E(\mathbf{X}) = \begin{pmatrix} E(X_1) \\ E(X_2) \\ \vdots \\ E(X_k) \end{pmatrix}, \quad V = Var(\mathbf{X}) = \begin{pmatrix} Var(X_1) & Cov(X_1, X_2) & \dots & Cov(X_1, X_k) \\ Cov(X_2, X_1) & Var(X_2) & \dots & Cov(X_2, X_k) \\ \vdots & \vdots & \ddots & \vdots \\ Cov(X_k, X_1) & Cov(X_k, X_2) & \dots & Var(X_k) \end{pmatrix}.$$

V is symmetric ($\exists$ a spectral decomposition).

For IID $X_1, \dots, X_k, V = \text{diag}(\sigma^2, \dots, \sigma^2)$.

$$\mathbf{X} \sim \mathcal{N}(\mu, \Sigma) \wedge V_{ij} = 0 \iff X_i, X_j \text{ are independent.}$$

3 Estimation of Population Mean

3.1 The Population

Gauss Model

A statistical model encompasses a set of statistical assumptions concerning the generation of sample data. There is no definitive method to check the assumption, but we assume data-generating protocols, or plot data over time for a systematic pattern.

The estimates are only as good as the statistical model.

We assume the **Gauss model**, such that $x_1, \dots, x_n$ are realisations from IID RVs $X_1, \dots, X_n$ with expectation μ and variance σ^2 .

Population

Consider a **population variable** denoted x , of which each of the N subjects in the population i has a fixed value x_i . x is **not a RV**. x has a distribution, comprised of its possible values and proportions. It can also be summarised using the **mean and variance** (often unknown):

$$\mu_1 := \frac{1}{N} \sum_{i=1}^N x_i, \quad \sigma^2 := \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = \mu_2 - \mu_1^2.$$

The unknown summary statistic is known as a **parameter**. Typically the parameter of interest is the mean μ .

We may also be interested in several popvars, of which we may approximate using a **multivariate distribution**. The **covariance** between popvars x and y is defined:

$$Cov(x, y) = \frac{1}{N} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y).$$

PopVars and RVs are **mathematically identical**, with proportion replaced by probability. However, there is no randomness, hence **probability does not apply**.

Making a random draw from a popvar of mean μ and variance

σ^2 , the x -value of the chosen subject X is a RV with the **same distribution** as that of the popvar.

$$E(X) = \mu, \quad Var(X) = \sigma^2.$$

In cases where the population is not apparent (**Coin-tossing**), we consider RV X , with $E(X)$ the variable of interest.

Continuous vs Discrete PopVars: Since x takes finitely many values ($\leq N$), there is **no clear distinction** between continuous and discrete popvars. Sometimes, it would simply be convenient to approximate with continuous distributions.

3.2 Estimating Population Parameters

Making n random draws **with replacement** from a population with popvar of mean μ and variance σ^2 . Let x_i be the i -th outcome of the draws. **Each x_i is a realisation of IID X_i** (replacement), and X_i has the same distribution as the popvar.

$$E(X_i) = \mu, \quad Var(X_i) = \sigma^2.$$

Estimating μ

We may consider the sample sum $s_n = \sum_{i=1}^n x_i$ and mean $\bar{x}$ **realisations** of the random sum and mean

$$S_n = \sum_{i=1}^n X_i, \quad E(S_n) = n\mu, \quad Var(S_n) = n\sigma^2.$$

$$\bar{X} = \frac{S_n}{n}, \quad E(\bar{X}) = \mu, \quad Var(\bar{X}) = \frac{\sigma^2}{n}.$$

We say the **estimate** $\bar{x}$ is a realisation of the **estimator** $\bar{X}$.

Standard Error

We measure the average size of error with the **standard error** in $\bar{x}$. Since $\bar{x}$ is a realisation of $\bar{X}$, on average $\bar{x}$ is about

$$SE := SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}$$

away from μ . By Jensen's inequality, $E(|\bar{X} - \mu|) < SD(\bar{X})$. SE is a **constant**.

Estimating SE – The Bootstrap

We estimate SE by estimating σ^2 with the **sample variance**, and then taking the square root. The sample variance s^2 is used

$$E(S^2) = E\left\{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right\} = \sigma^2$$

over the variance of the sample data v as

$$E(\hat{\sigma}^2) = \frac{n-1}{n} \sigma^2.$$

We only have $n-1$ realisations of $(X_i - \bar{X})^2$, if $n-1$ deviations are known, the last one is known as well as the deviations sum to zero.

The **estimate** SE is a realisation of the **estimator** $\frac{s}{\sqrt{n}}$. Note that **S is biased for σ** , but bias is small for large n .

Informally, we have

$$\mu \approx \bar{x} \pm \frac{s}{\sqrt{n}}.$$

Estimating p

In the special case of estimating μ where $X \sim \text{Bernoulli}(p)$, $\mu = p$ and $\sigma^2 = p(1-p)$. We have **the estimate $\hat{p}$ from the estimator $\hat{p} = \frac{S_n}{n}$** , and $SD(\hat{p})$ is estimated as $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$.

$$p \approx \hat{p} \pm \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

where $\hat{p}$ (here) is the realisation.

3.3 Estimation Bias

Consider a parameter θ estimated by estimator $\hat{\theta}$, based on IID RVs $X_1, \dots, X_n$. **MSE** is defined for an estimator $\hat{\theta}$ wrt to an unknown parameter θ as:

$$MSE := E\left\{\left(\hat{\theta} - \theta\right)^2\right\} = Bias^2 + SE^2.$$

where **$Bias := E(\hat{\theta}) - \theta$** . $\hat{\theta}$ is **unbiased** for θ if:

- $Bias = 0$
- $E(\hat{\theta}) = \theta$, or
- $MSE = SE^2$

Note: MSE has the same form for prediction of a RV with a constant.

Measurement Bias

We assume a modified **Gauss model**, such that $x_1, \dots, x_n$ are realisations from IID RVs of expectation $\mu + b$ and variance σ^2 , where b is a bias parameter.

$E(\bar{X}) = \mu + b$, $\bar{x}$ is a biased estimate of μ , and SE is a measure of how far $\bar{x}$ is from $\mu + b$. $MSE \rightarrow b^2$ as $n \rightarrow \infty$.

Bias cannot be reduced, or detected even from an infinite amount of measured data. Corrected by comparing to known standards.

3.4 Confidence Interval

We consider the construction of a confidence interval for μ . For $\bar{X}$ with **normal distribution and known σ^2** ,

$$P\left(-z_{\frac{\alpha}{2}} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha.$$

$$P\left(\mu - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \bar{X} \leq \mu + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

$$P\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

$$\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right)$$

is a $(1 - \alpha)$ -CI for μ .

Since $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim t_{n-1}$, for $\bar{X}$ with **normal distribution with unknown σ^2** ,

$$P\left(-t_{\frac{\alpha}{2}, n-1} \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq t_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha.$$

$$P\left(\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}\right) = 1 - \alpha.$$

$$\left(\bar{X} - t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}, \bar{X} + t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}}\right)$$

is a $(1 - \alpha)$ -CI for μ .

By CLT, for large n , $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$. For $\bar{X}$ with **any distribution with unknown σ^2** ,

$$P\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right) \approx 1 - \alpha.$$

$$P\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}\right) \approx 1 - \alpha.$$

$$\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}\right)$$

is an **approximate** $(1 - \alpha)$ -CI for μ .

For $X \sim \text{Bernoulli}(p)$, we have the special case $s =$

$$\sqrt{\hat{p}(1-\hat{p})}$$

$$\left(\bar{X} - z_{\frac{\alpha}{2}} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}, \bar{X} + z_{\frac{\alpha}{2}} \frac{\sqrt{\hat{p}(1-\hat{p})}}{\sqrt{n}}\right)$$

is an approximate $(1 - \alpha)$ -CI for p .

Interpretation

Suppose the experiment is **repeated many times**, each time taking a random sample and computing a $(1 - \alpha)$ %-CI. The CIs have the same length of different centres, **each a realisation of a random interval** containing μ with probability 95%. **Around** $(1 - \alpha)$ % **of the CIs contain μ** . We say that we are $(1 - \alpha)$ % **confident** that μ is between $\bar{X} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$ and $\bar{X} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$.

If the model is inaccurate, the level of confidence is less than $1 - \alpha$.

4 General Estimation

4.1 Family of Distributions

A **family of distributions** is a set of distributions that can be characterised by a **finite number of parameters**. The **parameter space** Θ is the set of all possible values of the parameters. If a bijective correspondence exists between Θ and the family of distributions, the parameters are **identifiable** (eg. unidentifiable when **predictors are collinear in regression**). Within the parameter space, we find the estimator $\hat{\theta} : \Omega \rightarrow \Theta$ of the parameter θ approximating the true distribution.

4.2 Method of Moments

The **k -th moment**, where $k \in \mathbb{N}^*$, of a random variable X is defined as $\mu_k := E[X^k]$. It is estimated by the **sample moment**

$$\hat{\mu}_k := \frac{1}{n} \sum_{i=1}^n X_i^k, \quad SD(\hat{\mu}_k) = \frac{\sqrt{\mu_{2k} - \mu_k^2}}{\sqrt{n}}$$

By equating the sample moments to the population moments and solving for the parameters, we obtain the **method of moments estimator** (MME)

$$\hat{\theta}_{MM} := g(\hat{\mu}_1, \hat{\mu}_2, \dots, \hat{\mu}_k)$$